BI-GY 7453: Prof. Manpreet S Katari
April 14, 2025

ASSIGNMENT 4: Due Date: 25 April 2025

File Format to submit- .ipynb with output mentioned in the file (No grades will be allocated if the file format is different or it lacks output)

Next-generation sequencing technologies are making a large impact in Biology today. The ability to sequence large genomes and RNA molecules from cells in different conditions has allowed biologists to propose many different experiments. At the root of all the analysis, the sequences have to be first aligned to the genome so you know where the sequence comes from.

For this assignment, you will parse the results of an alignment (called a SAM file) and compare it to a file that contains gene annotations (GTF) . The goal is to count the number of sequences in the SAM file that match each of the genes in the GTF file. The documentation for the SAM file can be found here.(https://samtools.github.io/hts-specs/SAMv1.pdf). The name of the SAM file is sample.sam. The two columns of interest for our purpose are column 3, the name of the chromosome, and column 4, the start position of the alignment. For simplicity of the homework, we are not worried about the strand and the length of the match.
An example of two lines from the File look something like this:

```
HANNIBAL_4_FC308YYAAXX:5:1:5:247 0 Chr3 14202145 60 50M * 0 0
CGGCCGGGGGACGGACTGGGAACGGCTCTCTCGGGAGCTTTCCCCGGGCG aaaa_]]aaaa`Z]aaaaaaaaaaaa__XXU]\[UGXKRZXPRMUOXVV[ AS:i: 0
XN:i:0 XM:i:0 XO:i:0 XG:i:0 NM:i:0 MD:Z:50 YT:Z:UU NH:i:1

HANNIBAL_4_FC308YYAAXX:5:1:5:527 0 Chr2 8023 0 46M4S * 0 0
AATGTAGGCAAGGGAAGTCGGCAAAATGGATCCGTAACTTCGGGAACTGT aaaaaaaaaa^X[__^X^a`_^^__`_[^_`__`X[X^`^\UGUOXU[^[ AS:i:-8 ZS:i:-8  XN:i:0
XM:i:0 XO:i:0 XG:i:0 NM:i:0 MD:Z:46 YT:Z:UU NH:i:2
```

Now the GTF file (Arabidopsis.gtf) is an already sorted tab-separated file. It contains the chromosome in the first column and the start and end positions of an exon in the 4th and 5th columns, and the gene name in the last column. To determine the number of reads that match a gene, count the number of start positions from the SAM file that fall between the start and stop coordinates of an exon that belongs to a gene.
Here is a quick look at the GTF file.

```
Chr1 TAIR10 exon 3631 3913 . + . transcript_id "AT1G01010.1"; gene_id "AT1G01010"; gene_name "AT1G01010"; Chr1 TAIR10 exon
3996 4276 . + . transcript_id "AT1G01010.1"; gene_id "AT1G01010"; gene_name "AT1G01010"; Chr1 TAIR10 exon 4486 4605 . + .
transcript_id "AT1G01010.1"; gene_id "AT1G01010"; gene_name "AT1G01010"; Chr1 TAIR10 exon 4706 5095 . + . transcript_id
"AT1G01010.1"; gene_id "AT1G01010"; gene_name "AT1G01010"; Chr1 TAIR10 exon 5174 5326 . + . transcript_id "AT1G01010.1";
gene_id "AT1G01010"; gene_name "AT1G01010"; Chr1 TAIR10 exon 5439 5899 . + . transcript_id "AT1G01010.1"; gene_id
"AT1G01010"; gene_name "AT1G01010"; Chr1 TAIR10 CDS 3760 3913 . + 0 transcript_id "AT1G01010.1"; gene_id "AT1G01010";
gene_name "AT1G01010";
Chr1 TAIR10 CDS 3996 4276 . + 2 transcript_id "AT1G01010.1"; gene_id "AT1G01010"; gene_name "AT1G01010"; Chr1 TAIR10 CDS
4486 4605 . + 0 transcript_id "AT1G01010.1"; gene_id "AT1G01010"; gene_name "AT1G01010"; Chr1 TAIR10 CDS 4706 5095 . + 0
transcript_id "AT1G01010.1"; gene_id "AT1G01010"; gene_name "AT1G01010"; Chr1 TAIR10 CDS 5174 5326 . + 0 transcript_id
"AT1G01010.1"; gene_id "AT1G01010"; gene_name "AT1G01010"; Chr1 TAIR10 CDS 5439 5630 . + 0 transcript_id "AT1G01010.1";
gene_id "AT1G01010"; gene_name "AT1G01010"; Chr1 TAIR10 exon 5928 6263 . - . transcript_id "AT1G01020.1"; gene_id
"AT1G01020"; gene_name "AT1G01020";
Chr1 TAIR10 exon 6437 7069 . - . transcript_id "AT1G01020.1"; gene_id "AT1G01020"; gene_name "AT1G01020"; Chr1 TAIR10 exon
7157 7232 . - . transcript_id "AT1G01020.1"; gene_id "AT1G01020"; gene_name "AT1G01020";
```

Notice that the third column defines the feature type. We are interested in the exons only, so ignore the CDS rows in the file. When you count the sequences that map to the exon, you should associate them with the gene_name provided in the last column. Many genes

have multiple exons.

Suggested strategy:

- First, split all the alignments in the SAM file based on chromosomes. Create a dictionary where the key is the chromosome name and the value is a list of start coordinates. Note that some lines do not have a value for chromosomes which means that sequence did not align to the genome, so we can ignore them.
- Next go through each exon in the GTF file and count the number of reads that are between the start and end coordinates. Since we know which chromosome the gene is in you should have to go through the start positions one at a time to see how many are within the start and end coordinate.

1) **Implement without sorting the SAM file**: Use the original sam file to determine the number of reads that match a gene, count the number of start positions from the SAM file that fall between the start and stop coordinates of an exon that belongs to a gene.
2) **Implement after sorting the SAM file**: For each chromosome, sort the alignment coordinates. You can use any of the functions discussed in class. Now again count the number of alignments that match a gene. ***Note*** Since the alignment coordinate is sorted, you don't have to check every single value in the list. Once your alignment coordinate is greater than the end coordinate of the exon, you can stop. Additionally, you can remove the alignment coordinates from the list because the genes are sorted so you will never have to look before the position you have already visited.

In both the above cases calculate the total time taken with sorting and without sorting the coordinates from the SAM file and clearly explain why sorting reduces/increases the total time for execution. You can use the below-mentioned code to calculate the total time taken to execute a function.

```
# in Python
import time
start = time.process_time()
# your code here
print(time.process_time() - start)
```